From Data to Insight: Identifying Key Drivers of Poor Mental Health Days

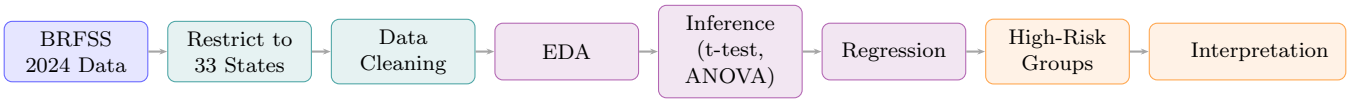
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Summary:-

Mental health is a major public health concern in the United States, influencing millions each year. Mental health can be influenced by many factors, including demographic factors (age, sex, income) and social or community factors such as loneliness, emotional support, life satisfaction and safe environment. Understanding which factors are most strongly associated with poor mental health days is important for identifying vulnerable groups and to help inform public health policy that may improve well-being. For this reason, this project aims to examine whether social determinants explain variation in poor mental health days beyond demographic factors and to identify key behavioral, physical health, and social variables associated with this outcome. It also seeks to identify high-risk population groups and evaluate whether physical activity is associated with fewer poor mental health days.

The analysis uses data from the 2024 Behavioral Risk Factor Surveillance System (BRFSS)[1], a nationwide survey conducted by the CDC. The dataset contains approximately 457,670 respondents and includes demographic, behavioral, physical health, and social variables, as well as poor mental health (measured using variable MENTHLTH)[1]. The variable records the number of days in the past 30 days when mental health was reported as not good. The project will use data from 33 U.S. states where social determinant questions were asked during the survey.

Proposed Plan:-



The analysis will begin with the preprocessing of the BRFSS dataset[1] (.xpt format). Special response codes will be converted to missing values (NA) using the codebook, and the dataset will be restricted to the selected states. Variables will be selected across four domains: demographic (age, sex, income), behavioral (physical activity, smoking), physical health, and social aspects (life satisfaction, emotional support, loneliness). Categorical variables will be recoded into labeled factors, and key variables will be checked for outliers and distribution patterns.

Exploratory data analysis (EDA) will use summary statistics and visualizations (histograms, bar charts, box plots) to examine the distribution of poor mental health days and differences across demographic, behavioral, and social determinant groups. Statistical methods will include two-sample t-tests for binary comparisons and one-way ANOVA for multi category variables (e.g., income, age, physical activity), along with confidence intervals to measure uncertainty in the estimates.

Multivariable linear regression will be used to evaluate the combined effects of predictors such as demographic, behavioral, physical health, and social variables on poor mental health days (MENTHLTH). It will also determine whether social factors contribute to explaining the variation in poor mental health days beyond demographic factors, and whether physical activity is associated with fewer poor mental health days. Possible challenge includes missing data in multiple variables. This will be addressed through data preprocessing, including the removal of invalid responses, imputation where appropriate, and analysis of the type of missingness in variables, such as MCAR, MAR, and MNAR.

References

[1] Centers for Disease Control and Prevention (CDC). *Behavioral Risk Factor Surveillance System (BRFSS)*, 2024. https://www.cdc.gov/brfss/annual_data/annual_2024.html